

ICS 604: Applied Data Science – Final Project Requirements

The final project provides an opportunity to apply the concepts, tools, and methods covered in this course to a self-selected data science problem. The project consists of three components (percentage of total course grade in parentheses):

- Project Proposal (7%)
- Peer Feedback on Proposals (8%)
- Final Report and Code Submission (20%)

Note: The project requirements have been updated from those originally described in the course syllabus. Instead of a proposal presentation, peer feedback on project proposals is now required. The syllabus on Lamaku has been updated to reflect this change.

1. Project Proposal (max 2 pages)

Due: Monday, March 23 (11:59 PM)

The proposal should clearly describe your planned project and demonstrate that it is both feasible and meaningful. It should include the following:

- **Motivation**
 - Why did you choose this problem or dataset(s)?
 - Why are the questions you aim to answer important or interesting?
 - What insights or outcomes do you expect to gain?
- **Dataset**
 - Brief description of the dataset(s), including source, size, and key variables/features.
 - Anticipated preprocessing or data-cleaning steps.
 - Any potential challengers (e.g., imbalance, missing data, limited sample size, etc.).
- **Research Questions**
 - Clearly state the main questions or hypotheses your project aims to address.
 - Ensure they are specific, focused, and measurable.
- **Methods / Approach**
 - Outline the analytical methods or approaches you plan to use.
 - Mention relevant models, algorithms, or statistical/data analysis techniques.
 - Briefly explain why these methods are appropriate for your research questions.

Note: Proposals will be shared with the class for peer feedback.

2. Peer Feedback on Project Proposals (approximately half a page per proposal)

Due: Monday, March 30 (11:59 PM)

Each student will provide concise and constructive feedback on *two assigned proposals*. This component also **contributes to participation credit**.

Your feedback should address the following points:

- **Clarity:** Are the research questions clearly stated, focused, and well-defined?

- **Feasibility:** Is the project scope reasonable given the time frame and available data?
- **Methodology:** Are the proposed methods appropriate for addressing the research questions?
- **Data Suitability:** Is the dataset(s) appropriate and sufficient for the proposed analysis?
- **Suggestions:** Provide at least 2–3 specific suggestions for improvement (e.g., refining research questions, alternative methods, potential risks, additional analyses, etc.).
- **Strengths:** Identify at least one strong aspect of the proposal.

Note: Feedback will be anonymized before being shared with the corresponding proposal author. Please ensure your comments are professional, constructive, and concise.

3. Final Report and Code Submission

Due: Monday, May 11 (11:59 PM)

The final submission must include both a written report and complete, reproducible code.

A. Written Report (3 – 4 pages)

The report should be clearly organized, concise, and well written. Include the following:

- **Introduction and Motivation**
Briefly describe the problem and explain its significance or relevance.
- **Dataset Description**
Describe the data source, key features, preprocessing steps, and any challenges encountered.
- **Research Questions / Objectives**
Clearly (re)state the questions or objectives addressed in your project.
- **Methods / Approaches**
Describe the models, algorithms, or analytical techniques used, and justify why these methods are appropriate for your problem.
- **Results and Analysis**
Present your findings using figures, tables, or visualizations, and clearly interpret and explain the results.
- **Discussion / Insights**
Discuss the significance of your findings, address limitations, and suggest possible improvements or extensions.
- **Use of AI Tools (if applicable)**
Specify any AI tools, models, or frameworks used, and clearly explain how they were applied in your project.
- **Conclusion**
Summarize the key takeaways and overall insights.

B. Code Submission

- Create a GitHub repository for your project. Make it public, or grant access to the instructor and TAs if it cannot be public.
- Ensure your code is well organized and clearly documented.
- Include instructions for running your code if needed (e.g., in a README or comments).
- Ensure all necessary files (code and data, if permitted) are included so that your results can be reproduced.